

Roy Cohen

Algorithm Engineer

CONTACT

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PROFESSIONAL EXPERIENCE

R&D Algorithm Engineer – Rafael Advanced Defense Systems

Apr 2025 – Present

- Develop navigation and mission planning algorithms for terrain-aware autonomous systems
- Implement graph search and combinatorial optimization solutions including A*, Dijkstra, TSP, and GTSP variants
- Design heuristics and cost functions for real-time route optimization under operational constraints
- Evaluate routing and optimization performance under varying operational constraints using simulation and operational research techniques
- Develop tools and evaluation pipelines using Python, C++, MATLAB, and Git

PROJECTS

Autonomous Traffic Violation Detection using Edge AI

- Built a real-time curb parking violation detection system on Jetson Orin Nano
- Designed end-to-end inference and event detection pipeline for embedded deployment using NVIDIA DeepStream

Wildfire Detection using Edge AI

- Developed a real-time wildfire detection pipeline on edge AI hardware using camera-based computer vision
- Optimized inference performance for low-power deployment scenarios
- Awarded 2nd place in the international SD Express Student Challenge, winning a total prize of \$10,000 and presenting in IMVC conference

Space Invaders Game on FPGA

Created a complete “space invaders” style game on FPGA

EDUCATION

Technion University

B.Sc. in Electrical Engineering, 4th year

- GPA: 85
- Took courses in Algorithms and Data Structures ,Machine Learning, VLSI, Signal Processing and Computer Architecture

SKILLS

- Languages: Python, C++, C, MATLAB
- Algorithms: A*, Dijkstra, TSP, GTSP, heuristic optimization
- Tools: Git, Linux, Jetson Orin Nano, Docker, DeepStream

ADDITIONAL

- Licensed skipper and active participant in the Technion sailing team
- Competitive endurance sports and outdoor expedition background, including the Israel National Trail